

MAKSYMILIAN KALEK

SENIOR FULL-STACK AI SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Innovative Senior Full-Stack AI Software Engineer with 5+ years of experience building production-grade Generative AI systems and cloud-native applications. Proven expertise in LLM integration, agentic orchestration, Retrieval-Augmented Generation (RAG), and end-to-end MLOps pipelines. Track record of delivering impactful AI solutions across healthcare, chemicals, semiconductor, and automotive industries - bridging the gap between cutting-edge technical architectures and strategic business goals to drive measurable outcomes.

CORE COMPETENCIES & TECHNOLOGIES

AI / ML / NLP	LLMs, Agentic Development, LangChain, LangGraph, RAG, Hugging Face, Transformers, PyTorch, OpenAI API (Azure/GPT), Databricks, OCR
Cloud (AWS)	AWS CDK, Lambda, Bedrock, API Gateway, DynamoDB, S3, SQS, ECS/EKS, Cognito, IAM, Serverless Framework
Backend	Python (FastAPI, Flask), Java 17/19 (Spring Boot, Spring MVC, Hibernate), Microservices, RESTful APIs
Frontend	React, TypeScript, JavaScript, MUI, HTML/CSS, Cypress, Jest, React Testing Library, Tailwind
Devops / MLOps	Docker, Kubernetes, Kubeflow, MLFlow, CI/CD (GitHub Actions, Jenkins), Git, Bash, Agile/Scrum

WORK EXPERIENCE

AI Engineer | **Connecty AI** October 2025 - March 2026

- Spearheaded the architecture design and engineering of a large-scale AI data processing and business intelligence platform handling complex, high-throughput analytics.
- Drove agentic development initiatives, building autonomous LLM-powered agents to automate decision-making workflows across advertising channels.
- Designed and implemented advanced agent orchestration frameworks ensuring seamless communication and task execution across multiple AI models and microservices.
- Developed LLM prompt serialization systems for advertisement decision engines, including domain instruction blocks and structured output pipelines.

Full-Stack AI Software Engineer | **Perelyn (AI)** August 2023 - September 2025

- Led technical AI teams and owned end-to-end delivery of complex LLM and cloud projects for diverse enterprise clients across multiple industries.
- Mentored junior developers, moderated technical workshops, conducted rigorous code reviews, and presented architectural strategies to senior leadership.
- Key Projects:
 - Enterprise AI Data Processing | Chemicals:** Engineered a high-performance backend platform using LangChain, LangGraph, and OCR to process and sanitize ~100,000 technical PDF documents. Deployed cost-optimized serverless AWS infrastructure (CDK, Lambda, DynamoDB).
 - MLOps Infrastructure | Global Semiconductor Enterprise:** Architected and deployed a production-grade MLOps ecosystem using Kubernetes, Kubeflow, and MLFlow to standardize and scale machine learning workflows across a global engineering organization.

Continues below ↓

- **AI Healthcare Platform | Adiu Health (adiu-health.de):** Built a full-stack LLM-powered healthcare documentation solution from scratch. Integrated Azure OpenAI and AWS Bedrock, developed RAG infrastructure, set up CI/CD pipelines, and delivered a React-based frontend in collaboration with UX/UI designers.
- **AI Consulting & CV Optimization:** Engineered a Gen-AI platform providing automated, tailored CV adjustments and AI-driven consulting matching. Leveraged LangChain and advanced prompting pipelines to process user data and match consultants to potential projects, supported by a scalable AWS architecture.

Full-Stack Software Engineer | Grape Up (Automotive)

September 2022 - July 2023

- Developed a comprehensive fleet vehicle monitoring hub for a major car rental company using Java 19, Spring Boot, and Python.
- Designed highly scalable, cloud-based serverless solutions using AWS CDK, Lambda, and DynamoDB.
- Built responsive frontend interfaces in React/TypeScript with high test coverage via Cypress and Jest.

Java Software Engineer | SIX (Banking)

January 2022 - August 2022

- Built robust data processing pipelines to extract and analyze data from thousands of ATMs, streamlining maintenance for a global financial infrastructure provider.
- Engineered microservice architectures using Java 17, Spring, Hibernate, and SQL. Optimized CI/CD workflows with Jenkins and Docker.

Data Scientist | I2G (Data) part-time

January 2022 - March 2022

- Developed a Python analytics platform (Pandas, NumPy, Scikit-learn) to classify characteristics from large photo book datasets.

Java Software Developer | Best.net (E-Commerce)

July 2020 - December 2021

- Contributed to a large-scale B2B/B2C e-commerce platform using Java 8/11, Apache Struts, and PostgreSQL. Ensured reliability with Selenium end-to-end tests.

EDUCATION

B.Sc. Informatics and Econometrics (Specialization: Data Science)

University of Economics and Business in Poznań (2019-2022)

CERTIFICATIONS

- LLM101x: Large Language Models - Application through Production (edX / Databricks)
- LLM102x: Large Language Models - Foundation Models from the Ground Up (edX / Databricks)

LANGUAGES

- English - C1
- German - A2
- Polish - Native